

Ethics In Research

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- **Research Deontology:** Research Malpractice
- **Bioethics:** Ethical Aspects of Research with Living Beings

Why Scientific Malpractice Happens?

Research Goal

The goal of Research is to satisfy human curiosity and to make the world a better place

Scientific attitude

- Critical (willing to be fair and objective)
- Open (willing to accept critics)
- Transparent (willing to socialized your ideas)

This is the ideal world!

Why Scientific Malpractice Happens?

In the real world we also have,

Human nature

Many scientists have big egos

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Publish or Perish

Scientists have a strong pressure from their institutions for getting good results and **publish** them as scientific papers in prestigious journals

- promotion
- funding
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- Scientists are tempted to do **scientific malpractice**
 - Is this an ethical dilemma? No. It is about ethical standards

Deontological approach: let's try to identify the ethical norms of research

Is Scientific Malpractice Really a problem?

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Retractions

Retraction= a paper is **symbolically** eliminated

- 10000 retractions in 2024 and the number keeps increasing
- The countries with the most retractions were Saudi Arabia, Pakistan, Russia, and China.

According to *Retraction Watch*:

- 2% of papers should be retracted because fraud or severe bad mistake
- 0.1% of papers are retracted

Mendel (1822-1884): A controversial case



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If he really manipulated data, what would be your moral judgment?

[Go here for more details](#)

Malpractice: (unacceptable)

- Fraud
- Plagiarism
- Conflict of Interest

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Questionable Research Practice: (unacceptable, but harder to characterize, identify and control)

- Undue Influence of personal values
- Harking
- *p*-hacking
- Cherry picking
- Selective Omission
- Data Snooping

Fraud

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 - It may mislead other researchers
 - wasted resources (mislead funding agencies)
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Example (Wakefield scandal)

Related the development of autism and vaccines (using fraudulent data).
It has had an incredibly negative impact on public health.
It has messed with good parenting.
It has wasted a lot of research time and money.

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 - *Guidelines for Making Empirical AI Research Reproducible*
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Problem: Private companies' Research keeps secrets

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Solution:

- Transparency (e.g. open access journals)
- Automatic tools
- Publish in reputable journals and conferences

Undisclosed Conflict of Interests

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Solution:

- Note that what is wrong is **not to disclose** the conflict
- Transparency: Acknowledgements

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Example

- Psychology: I am against gay marriage, so I look for evidence of children of gay couples having psychological problems
- Comp. Science: I favor free software, so I look for evidence that Windows is worse than Linux.

This is difficult to solve, because you cannot ask researchers to disclose their personal values

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Example

Suppose that we are studying teacher's *quality* at UPC and our hypothesis is that CS teachers are better than Math teachers.

We use student surveys as a source of information.

Should we remove data from students that did not attend a minimum number of lectures if it favors our hypothesis?

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Solution:

- Report the manipulation
- If possible, pre-decide (and pre-register) the manipulation rules

Questionable Research Practice

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Example: Decide that a patient has fever at a given threshold. How serious is the problem?

- Historically, a p -value of 0.05 or lower is what is considered valid
- The number of studies in which the p -value is very close to 0.05 is surprisingly high!
- That is a sign of p -hacking being frequent

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Ex: Chose instances out of standard benchmarks

Solution: Tell the truth (*experiments demonstrate that idea may work well on some relevant instances*)

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Data Snooping: Stop the data collection when results are favorable

Ex: Time cutout with anytime algorithms

Solution: Tell the truth (*with longer cutouts the advantage of the algorithm does not show so clearly*)

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Example

Imagine I have access to Math exams scores at UPC in the last 10 years and I observe that women score significantly better than men.

I write the following paper: *i)* My hypothesis is that women are better than man at math because blah blah blah, *ii)* let's check my hypothesis on UPC exams of last 10 years, *iii)* my hypothesis is supported by the data, so the hypothesis seems to be correct

The European Code of Conduct for Research Integrity (All European Academies, ALLEA)

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 - For instance, in ancient times, doctors could not open dead bodies to investigate, or painkillers were not moral
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 - For instance, in ancient times, doctors could not open dead bodies to investigate, or painkillers were not moral
- **Kantian foundations:** all humans deserve the same consideration and respect
- Now extended to animals

An experiment using HUMANS is acceptable if,

The Four Fundamental Principles of Bioethics:

- ① Do not harm
- ② Do good
- ③ Respect human autonomy
- ④ Be fair with all humans

Do not harm (a.k.a. non-maleficence)

Nonmaleficence

Obligation of not to harm the subjects.

- do not kill,
- do not cause pain or suffering,
- do not incapacitate,
- do not cause offense,
- do not deprive others of the goods of life

Do Good (a.k.a. beneficence)

Beneficence

Obligation to act for the benefit of society.

- recognizes that researchers are altruistic and in debt to society

Autonomy

Subjects have the power to make rational decisions and moral choices, and each should be allowed to exercise his or her capacity for self-determination

- informed consent
- truth-telling
- confidentiality

Justice

Fair, equitable, and appropriate treatment of persons

- Regarding subjects of the study
- Regarding beneficiaries of the research

Conflicts between Principles

Conflicts

Each one of the 4 principles of ethics is to be taken as a *prima facie* obligation that must be fulfilled, unless it conflicts, in a specific instance, with another principle.

When faced with such a conflict, the researcher has to determine the actual obligation to the subjects by examining the respective weights of the competing obligations based on both content and context.

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Note that the 4 principles fit into the **Deontological** perspective (rules), but conflict resolution requires to incorporate an **Utilitarian** perspective (evaluate consequences of different choices).

Four Principles: example

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- ④ **(Be fair)** The drug must eventually be made available to everybody

Bioethics during Covid

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 - *Challenge studies*: deliberately infecting volunteers
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- During Covid pandemic some people (including scientists) asked for taking shortcuts in the process of developing vaccines
 - *Challenge studies*: deliberately infecting volunteers
- Those people gave utilitarian argumentations for their request
- They did not succeed because the **Do not harm** principle was not sufficiently clear while the **Do good** principle was uncertain
 - Infected people could die

Example: Experimenting with Brain-dead humans

Real Case

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Real Case

- Organs shortage for transplant is a serious health problem
- Pig-grown organs is the most promising solution
- Experimenting with Brain-dead humans to test pig-grown organs in humans
- Confront it with the 4 principles(do not harm and respect autonomy are problematic)
- Do not harm the brain-dead human is easy to justify, but what about the family?
- Autonomy is transferred to relatives
 - Respect family pain
 - Ask family what the patient would have wanted
 - Truth-telling and confidentiality becomes super important

[find details here](#) (Unexplainable podcast 19/12/2024)

The podcast explains how an Ethical Committee dealt with a real case.

FUTURE PERFECT

A bioethics professor weighs in on the Last of Us finale

Is it ever justified to kill one person to save many others?

by **Oshan Jarow**

Mar 17, 2023, 2:00 PM GMT+1



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Experiments with non-humans



- **Research and Drug testing with animals:**

- 11.5 million animals were used in experiments across Europe in 2011 (mostly for basic research)
- Some experiments are quite cruel
- It is known that most animals used in experiments are sentient (suffer).
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- Legislation aims at forbidding unethical behaviors
- Confront it with the 4 principles of Bioethics: not very useful.

Do Animals Deserve Ethics?

- The first question to be addressed is where we place the moral value of humans in relation with the moral value of the rest of animals
- Recall that Ethical Theories were proposed at a time when animals were not considered to have dignity or feelings or intelligence...
- ...so we may need to update them to our current moral framework

- Animalist Movements give the **same moral value to animals and humans** because,
 - Animals are **sentient** (capacity to have feelings)
 - stress (depression)
 - Animals are **intelligent** (at least they have sophisticated cognitive concepts and awareness)
 - Primates
 - Octopus
 - (New Caledonian) Crows

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- **Pete Singer** (Australia, 1946) makes a **utilitarian** approach and says that the **feelings of animals have the same value** as the feelings of humans
 - Singer claims that it is morally permissible to euthanise newborn children with severe disabilities (his life only causes suffering to the family)

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- **Pete Singer** (Australia, 1946) makes a **utilitarian** approach and says that the **feelings of animals have the same value** as the feelings of humans
 - Singer claims that it is morally permissible to euthanise newborn children with severe disabilities (his life only causes suffering to the family)
- **Tom Regan** (USA, 1938) makes a **deontological** approach and says that **animals have the right** to be treated with respect and dignity (not to be used as a mean)
 - Who would you save first: a baby or a dog? if the baby is mentally retarded Regan claims that he would save the dog

Experimenting with animals: Utilitarian approach

Moral balance sheet:

- Good consequences (*hedons*)
 - What is the goal of research?
 - How likely it will succeed?
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 - How painful is the disease in humans?
 - Which animals and how many?
 - How cruel is the experiment?

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Also it seems important to decide if we consider animal suffering of less importance as we go down in the evolution line (chimp vs mice vs fly)

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 - Experimenting with animals clearly treats animals as a mean

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So the discussion here is about if **all animals** deserve respect

Virtue ethics and animal rights

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Therefore, a virtuous person cannot ignore the mistreatment of animals.

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Virtue Ethics:

- Which are the virtues in this case?
 - compassion (to who?), progress enthusiasm, efficiency,...
- What would a virtuous person do?
 - probably here being virtuous means finding a good balance

Experiments with Animals: a pragmatcal trade-off

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Principle of the three Rs: (Russell and Burch, 1959)

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- **Reduce** (less animals)
- **Refine** (less pain)

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But there are other aspects also to be considered: What about frivolous research? (e.g. cosmetics), What about bad-quality research? (e.g. research as marketing)

An **Independent Ethical Board** must approve the research

Experimenting with animals. Another dilemma

Research with primates, (The Intelligence 1/9/21)

- Europe is putting more control on primates experiments
- Asia is making it easier
- Asia will have scientific advantage
- What to do?
 - Revert policy
 - Keep policy but benefit from knowledge
 - ...

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We can apply Kant's Universality principles to Countries

Back to experiments with Humans: The Polio Vaccine case

Polio Vaccine, developed in the 50's

- 1 Developed by Salk in the US
- 2 First tests on thousands of monkeys
- 3 Then, he injected himself, his wife and his three children
- 4 Finally, 1.8 million American kids
- 5 He never made a patent

This story took place before current protocols for drug development. It is a nice story of passion and generosity, but the third item would not be acceptable today and the fifth item seems impossible today

- third item is problematic w.r.t. the *respect human autonomy* principle of bioethics
- fifth item fits very well the *be fair* principle of bioethics but we are not that generous in these days

Experiments with humans: The Tuskegee case

- Tuskegee syphilis experiment (1932-1972)



Experiments with humans: The Tuskegee case

- Tuskegee, Mississippi (1932-1972)
- Goal: study untreated syphilis
- Subjects were poor, black, mostly illiterate people from the farmlands.
- they were told that they were offered free health for *bad blood* (a disease that does not exist) and given placebo.
- infected people were never informed (syphilis has many consequences, it can be propagated)
- they were never treated (antibiotics heal it)
- When it went public, big scandal
- Protocols for ethical drug testing on humans are now very well established

Informed Consent

if any research is conducted on a **human** subject, then this human subject must be **informed** about the experiment, must **consent** to the experiment voluntarily, without any coercion, and must have the right to **withdraw** consent at any time.

Informed Consent Related Problems

- Sometimes complete information is not possible (e.g. psychology).
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- Ethical issues about asking for consent:
 - when it is asked? early request, before medical treatment
 - how it is asked? fine print
 - what is the data going to be used for? clarity about the research

Independent Ethical Committees

Those conducting the research may be biased about the research design so there are,

Ethical Committees

Have an independent board approving and supervising the study

- **USA:** Independent Review Boards (**IRBs**)
- **Europe:** Research Ethics Committees (**RECs**)
- **Spain:** **Current legislation**
- **UPC:** **Ethics Committee**

- **Biomedicine:** there is a precise and consolidated legislation (Spain, EU,...)
- **Social Science:** there is no legislation. Only ethical recommendations (EU, last version 2021)

Experiments with humans: Computer Science

- Done in many domains:
 - Robotics (assistance to elderly, virtual surgery, ...)
 - Diagnosis (Image processing,...)
 - Behaviour Psychology (Human Machine Interaction, ...)

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 - Behaviour Psychology (Human Machine Interaction, ...)
- Some recent scandals have raised social alert
 - (The Guardian) *Facebook fiasco: was Cornell's study of emotional contagion an ethics breach?*
 - (The Times) *Google dreams up future of manipulating everyone*
 - Cambridge Analytics (in the Philippines)

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 - Diagnosis (Image processing,...)
 - Behaviour Psychology (Human Machine Interaction, ...)
- Some recent scandals have raised social alert
 - (The Guardian) *Facebook fiasco: was Cornell's study of emotional contagion an ethics breach?*
 - (The Times) *Google dreams up future of manipulating everyone*
 - Cambridge Analytics (in the Philippines)

Ethics in Computer Science Research?

May be we need a *Tuskegee* scandal to raise social pressure

Have we Had our Tuskegee?

The Facebook's Whistleblower Frances Haugen

- Mark Zuckerberg has unilateral control over 3bn people
- One Facebook study found that 13.5% of U.K. teen girls in one survey say their suicidal thoughts became more frequent after starting on Instagram.
- Another study found 17% of teen girls say their eating disorders got worse after using Instagram.
- About 32% of teen girls said that when they felt bad about their bodies, Instagram made them feel worse,

Neuroscience, Vol 397, 1/15/2019, Pages 31-40

Bilateral Prefrontal Cortex Anodal tDCS Effects on Self-reported Aggressiveness in Imprisoned Violent Offenders

- **tDCS**: placing electrodes on inmates foreheads and sending a current into their brains.
- The electricity will target the **prefrontal cortex**, a brain region that plays a role in **decision-making** and **social behavior**.
- The idea to be tested is that stimulating more activity in that region may make the prisoners **less aggressive**
- If the hypothesis is true, there are many uses for this new knowledge.

Study Case: In favor

- Procedure is painless
- Inmates give informed consent
- Incarceration is immoral and counterproductive
- Prison life carries out a lot of suffering
- If study succeeds: massive inmate release?
- The study may have many potential future benefits

Study Case: against

- Consent is given in a coercive situation (they may get benefits if they show good behaviour)
- the treatment could erode our free will



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Journal of Business Research



Customer churn prediction in the online gambling industry: The beneficial effect of ensemble learning



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ABSTRACT

The online gambling industry is one of the most revenue generating branches of the entertainment business, resulting in fierce competition and saturated markets. Therefore it is essential to efficiently retain gamblers. Churn prediction is a promising new alternative in customer relationship management (CRM) to analyze customer retention. It is the process of identifying gamblers with a high probability to leave the company based on their past behavior. This study investigates whether churn prediction is a valuable option in the CRM palette of the online gambling companies. Using real-life data of poker players at bwin, single algorithms, CART decision trees and generalized additive models are benchmarked to their ensemble counterparts, random forests and GAMens. The results show that churn prediction is a valuable strategy to identify and profile those customers at risk. Furthermore, the performance of the ensembles is more robust and better than the single models.

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abstract

The online gambling industry is one of the most revenue generating branches of the entertainment business, resulting in fierce competition and saturated markets. Therefore **it is essential to efficiently retain gamblers**. Churn prediction is a promising new alternative in customer relationship management (CRM) to analyze customer retention. It is the process of identifying gamblers with a high probability to leave the company based on their past behavior. This study investigates whether churn prediction is a valuable option in the CRM palette of the online gambling companies. Using real-life data of poker players at bwin, single algorithms, CART decision trees and generalized additive models are benchmarked to their ensemble counterparts, random forests and GAMens. The results show that churn prediction is a valuable strategy to identify and profile those customers at risk. Furthermore, the performance of the ensembles is more robust and better than the single models.